# Adaptive interrupt coalescing patch for NVMe SSD

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# What is Adaptive Interrupt Coalescing

- Heavy small I/O stress on multiple NVMe SSDs can trigger interrupt bursts, possibly up bounding IOPS of those NVMe SSDs with high level of HW spec.

- To solve interrupt burst:
  - If io size <= 32k, queue is busy and busy queues number at the same time larger than threshold(default is 2, can be modified during runtime), automatically enable coalescing interrupt.
  - If io size <=32k, queue is not busy, and busy queues number at the same time less or equal than threshold(default is 2, can be modified during runtime), automatically disable coalescing interrupt.



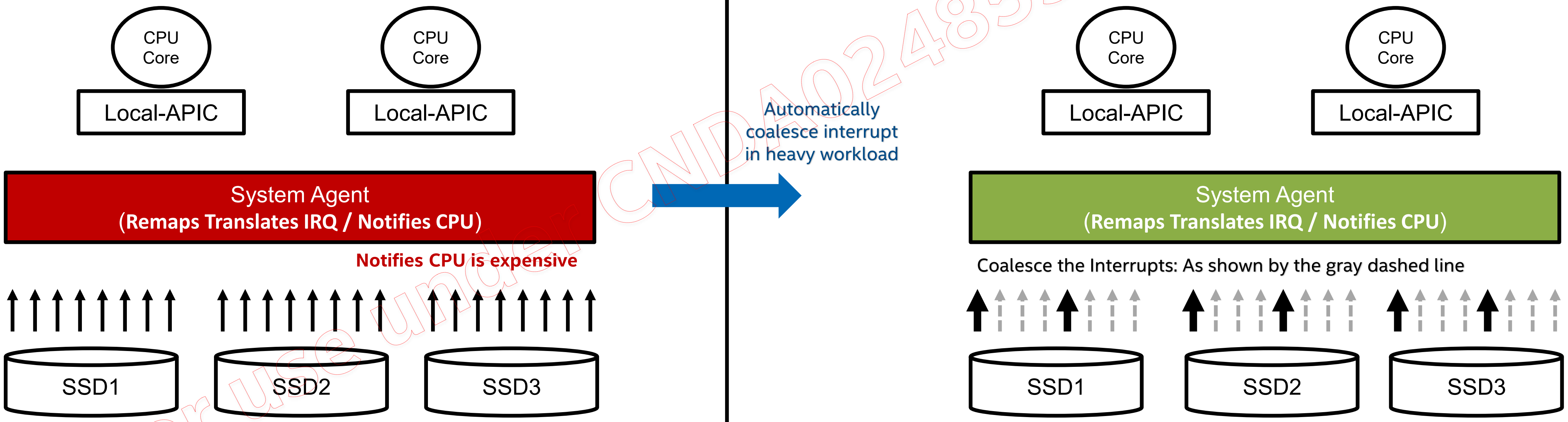
Note: no interrupt coalescing setting for io size larger than 32k

# How the patch works in Heavy WL

## Heavy Workload Problem

(e.g. bs=4k, numjobs=8 iodepth=128)

## How the Patch Works – Interrupt Coalescing Enabled

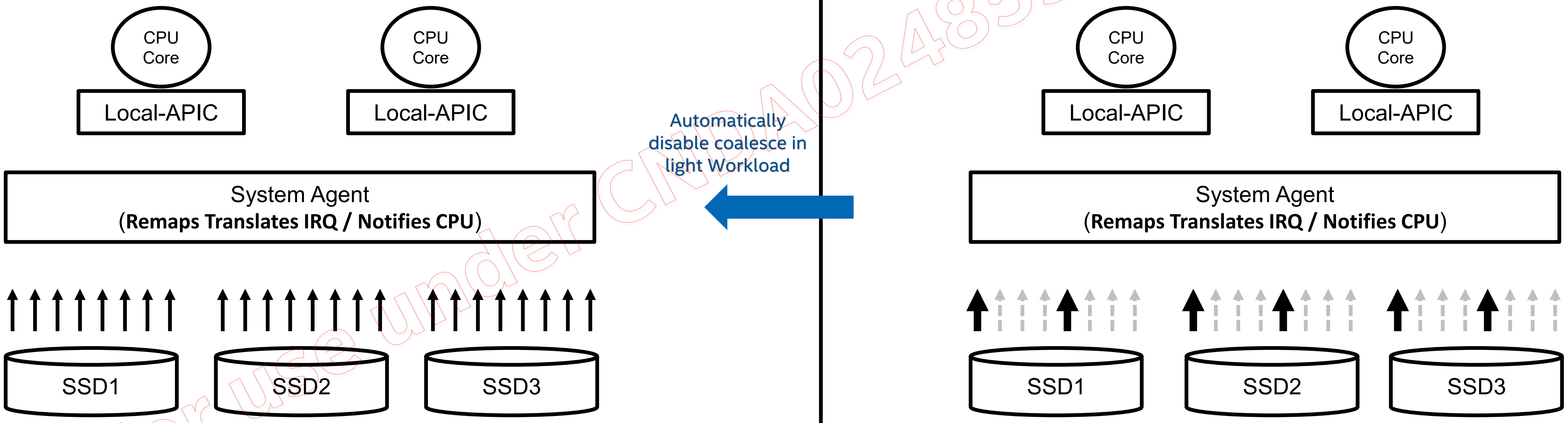



✓ Interrupt handling details: https://lwn.net/Articles/951186/

# How the patch works in Light WL

## How the Patch Works – Interrupt Coalescing Disabled

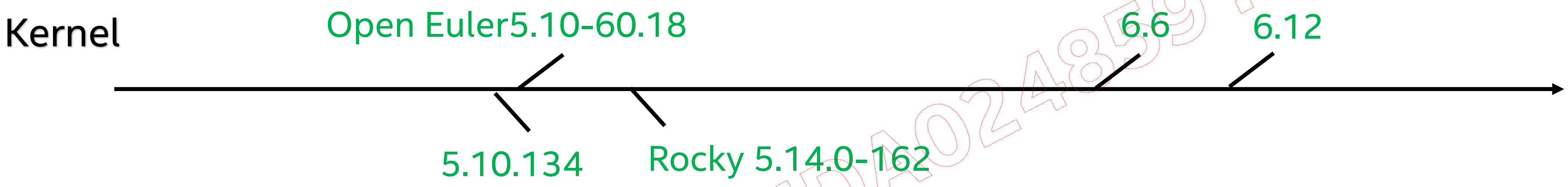
## Light WL
(bs=4k numjobs=1 iodepth=1)




Why low: (1) wait for requests (2) min timeout: 100us

✓ Interrupt handling details: https://lwn.net/Articles/951186/

# Patches Readiness




Patch for Green version is ready

# Patch Major Changes

- 描述下Patch的代码的主要改动

1. check workload on each running nvme queue, when meet some condition setting/clearing interrupt coalescing automatically;

1) check the inflight io num to judge whether setting or clearing interrupt coalescing

2) check completed io num during the judgement of setting or clearing interrupt coalescing;

2. enable interrupt coalescing for write with io size less than 32k, no interrupt coalescing for larger io size;

3. record the nvme queues been used currently and only enable interrupt coalescing when there are more than 2 nvme queues been used (can be modified during runtime);

4. add debug variable to print the io info of used nvme queues.

# Installation

## ❑ Steps to install

- git clone related Linux kernel source code
- git apply the patch
- compile Linux kernel
- make install Linux kernel
- reboot OS

## ❑ How to check if installed successfully

ls –l /sys/kernel/nvme/

If the configurations exists in this folder, it means the patch has been successfully installed.

```
[root@sh14l02SHFs0402 k5.14]# ll /sys/kernel/nvme/
total 0
-rw-r--r--. 1 root root 4096 Oct 15 15:05 coalesce_value
-rw-r--r--. 1 root root 4096 Oct 15 15:05 debug
-rw-r--r--. 1 root root 4096 Oct 15 15:05 disable
-rw-r--r--. 1 root root 4096 Oct 15 15:05 hit_count_before_change
-rw-r--r--. 1 root root 4096 Oct 15 15:05 max_coalesce_threshold
-rw-r--r--. 1 root root 4096 Oct 15 15:05 min_coalesce_threshold
-rw-r--r--. 1 root root 4096 Oct 15 15:05 min_completed_threshold
-rw-r--r--. 1 root root 4096 Oct 15 15:05 min_running_queues
-rw-r--r--. 1 root root 4096 Oct 15 15:05 switch_cnt
```

# Configuration

*Don't change the value before you really know what you are doing.

- ❑ configurations are located in /sys/kernel/nvme/

| Parameters | Default | Descriptions |
|---|---|---|
| coalesce_value | 266 | The coalesce value set to NVME SSD, similar to below command:<br>*nvme set-feature /dev/nvme$i -f 0x8 --value=266 -n 0*<br>266 == 0x10a, that means timeout is set to 100us, coalesced request count is set to 0xa.<br>If want to set 0x109, just need to "echo 265 > /sys/kernel/nvme/coalesce_value" |
| debug | 0 | 0: not print, "echo 0 > /sys/kernel/nvme/debug"<br>1: print some debug print about each running nvme device queue with fix interval, "echo 1 > /sys/kernel/nvme/debug" |
| disable | 0 | 0: enable, to enable adaptive interrupt coalescing, "echo 0 > /sys/kernel/nvme/disable"<br>1: disable, to disable adaptive interrupt coalescing, "echo 1 > /sys/kernel/nvme/disable" |
| hit_count_before_change | 5 | How many consecutive detection cycles should be satisfied before changing coalescing status between coalescing and non-coalescing. See more description in max_coalesce_threshold |
| max_coalesce_threshold | 8 | 8 op as a detection cycle, if there are over **8** unfinished requests in a NVME SSD queue in hit_count_before_change consecutive detection cycles, this patch will enable coalescing interrupt for that NVME SSD if not enabled yet.<br>Smaller value: more easily to enable coalescing, but may increase the status changes between coalescing enabling and disabling.<br>Bigger value: more difficult to enable coalescing<br>This value should be > min_coalesce_threshold. |
| min_coalesce_threshold | 2 | 8 op as a detection cycle, if there are less than **2** unfinished requests in a NVME SSD queue in hit_count_before_change consecutive detection cycles, this patch will disable coalescing interrupt for that NVME SSD if not disabled yet.<br>Smaller value: more difficult to disable coalescing. If <=1, will not disable coalescing.<br>Bigger value: more easily to disable coalescing, may increase the status changes between coalescing enabling and disabling.<br>This value should be >=2. |
| min_completed_threshold | 9 | another check condition to judge whether a nvme queue is busy or not, if the completion num during a detection cycle (8 new reqs) larger than this value, then we also treat this nvme queue as busy, if completion num during a detection cycle (8 new reqs) less than this value, then we also treat this nvme queue as idle. |
| min_running_queues | 2 | The above detection and judgement is nvme queue based, when a queue is treated as busy, also check whether there are other queues busy at the same time (less than 1s), if the busy queues num larger than this value, then set the interrupt coalescing. |
| switch_cnt | 0 | Indicate how many time the status has switched between coalescing and non-coalescing.<br>echo 1 > disable will reset this value to 0. |

# Patch with 16x UMIS UH812a PCIe Gen5 Reference Performance Data

| 容量 | 规格/Intel平台16盘实测 | 随机读（4k)(QD=1,job=1) | | 随机读（4k)(QD=32,job=1) | | 随机写（4k)(QD=1,job=1) | | 随机写（4k)(QD=32,job=1) | | 随机混合读写（4k)(QD=16,job=1) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 99% | 99.99% | 99% | 99.99% | 99% | 99.99% | 99% | 99.99% | r_99.9% | w_99.9% |
| 7680G | 单盘规格 | 85 | 85 | 150 | 210 | 10 | 20 | 160 | 210 | 400 | 400 |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| | kernel | | | | | | | | | | |
| | Intel平台16盘实测-1 | 61 | 72.81 | 95.5 | 140.62 | 5.29 | 16.38 | 74.69 | 88.75 | 293.56 | 16.35 |
| | Intel平台16盘实测-2 | 61 | 72.75 | 95.44 | 140 | 5.29 | 16.41 | 68.88 | 79.56 | 292.88 | 16.42 |
| | Intel平台16盘实测-3 | 61 | 72.88 | 95.31 | 140 | 5.29 | 16.16 | 64.44 | 78.81 | 291.06 | 16.42 |
| | 3轮平均值 | 61 | 72.81333333 | 95.41666667 | 140.2066667 | 5.29 | 16.31666667 | 69.33666667 | 82.37333333 | 292.5 | 16.39666667 |
| | 3轮平均值-规格达成 | 139.34% | 116.74% | 157.21% | 149.78% | 189.04% | 122.57% | 230.76% | 254.94% | 136.75% | 2439.52% |

测试FW版本：1082, FIO: thread=0

2路 GNR-SP 64C, HT on, Kernel: 6.6 + 自适应中断聚合Patch, FIO 配合numactrl --localalloc + 绑核 (FIO cpus_allowed参数)

| 容量 | 规格/Intel平台16盘实测 | IO场景 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | 顺序读（128k)-MB/s | 顺序写（128k)-MB/s | 随机读IOPS（4k)-k | 随机写IOPS（4k)-k | 随机混合读写IOPS（4k70读30写)-k | 随机读QD1时延（4k)-us | 随机写QD1时延（4k)-us | 顺序读QD1时延（4k)-us | 顺序写QD1时延（4k)-us |
| 7680G | 单盘规格 | 14800 | 10000 | 3100 | 410 | 696 | 55 | 9 | 8 | 9 |
| | 16盘规格 | 236800 | 160000 | 49600 | 6560 | 11136 | 55 | 9 | 8 | 9 |
| | | | | | | | | | | |
| | kernel | | | | | | | | | |
| | Intel平台16盘实测-1 | 238400 | 168700 | 55,005 | 7,547 | 19013 | 52.95 | 5.62 | 6.82 | 5.63 |
| | Intel平台16盘实测-2 | 238400 | 168700 | 54,985 | 7,536 | 18339 | 52.95 | 5.63 | 6.83 | 5.63 |
| | Intel平台16盘实测-3 | 238400 | 168700 | 54,978 | 7,356 | 18725 | 52.97 | 5.63 | 6.83 | 5.62 |
| | 3轮平均值 | 238400 | 168700 | 54989.33333 | 7479.666667 | 18692.33333 | 52.95666667 | 5.626666667 | 6.826666667 | 5.626666667 |
| | 3轮平均值-规格达成 | 100.68% | 105.44% | 110.87% | 114.02% | 167.86% | 103.86% | 159.95% | 117.19% | 159.95% |

测试FW版本：1082 每个盘片绑八个核，不跨node绑核 (numactl --localalloc)，thread=1，测试前先跑2遍4K随机写数据预埋

2路 GNR-SP 64C, HT on, Kernel: 6.6 + 自适应中断聚合Patch, FIO 配合numactrl --localalloc + 绑核 (FIO cpus_allowed参数)

数据可参见：https://www.intel.cn/content/www/cn/zh/artificial-intelligence/xeon-6-union-memory-enterprise-ssds.html

# Summary

**1. Adaptive Interrupt Coalescing Benefits:**

- Maintains high IOPS for heavy workloads.
- Ensures low latency for light workloads.

**2. Workload agnostic:**

- Workloads are agnostic to adaptive interrupt coalescing mechanisms in the kernel.
- Tools like htop and perf top indicate minimal kernel overhead.

**3. No Regression Observed:**

- Sequential 128k read/write (include case with io size larger than 32k) performance remains unaffected with adaptive interrupt coalescing.

# NextSetp

1. **Try optimization about this solution(e.g., do judgement with statistics from the whole NVMe device)**
2. **Target for upstream**